THE NORMALIZATION SPLITTING OF SIMPLICIAL OBJECTS IN AN ABELIAN CATEGORY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: rides on registered Lemma lemma-splitting-simplicial-groups; Mor_A(A, -) preserves kernels by the universal property of the kernel; the isomorphism of Equation equation-splitting is applied levelwise.

Editorial note. The label lemma-splitting-abelian-category names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Splitting of simplicial groups. Same unit, Lemma lemma-splitting-simplicial-groups. Each simplicial group Mor_A(A, U) is canonically split by Lemma lemma-splitting-simplicial-groups; the statement transports along these splittings.

Source reference. Original equation 017Q: sources/stacks-simplicial/pinned-original-source.tex, source lines 1990–1990. Prerequisite pointer invoked at proof line(s) 2267; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 017U: sources/stacks-simplicial/pinned-original-source.tex, source lines 2097–2097. Prerequisite pointer invoked at proof line(s) 2261; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let U be a simplicial object in $\mathcal{A}$. Then U has a splitting obtained by taking $N(U_0) = U_0$ and for $m \geq 1$ taking*

$$N(U_m) = \bigcap\nolimits_{i=0}^{m-1} \mathrm{Ker}(d_i^m).$$

Moreover, this splitting is functorial on the category of simplicial objects of $\mathcal{A}$.

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. For any object A of $\mathcal{A}$ we obtain a simplicial abelian group $\mathrm{Mor}_{\mathcal{A}}(A, U)$. Each of these are canonically split by Lemma 017U. Moreover,

$$N(\mathrm{Mor}_{\mathcal{A}}(A, U_m)) = \bigcap\nolimits_{i=0}^{m-1} \mathrm{Ker}(d_i^m) = \mathrm{Mor}_{\mathcal{A}}(A, N(U_m)).$$

Hence we see that the morphism (017Q) becomes an isomorphism after applying the functor $\mathrm{Mor}_{\mathcal{A}}(A, -)$ for any object of $\mathcal{A}$. Hence it is an isomorphism by the Yoneda lemma. $\square$